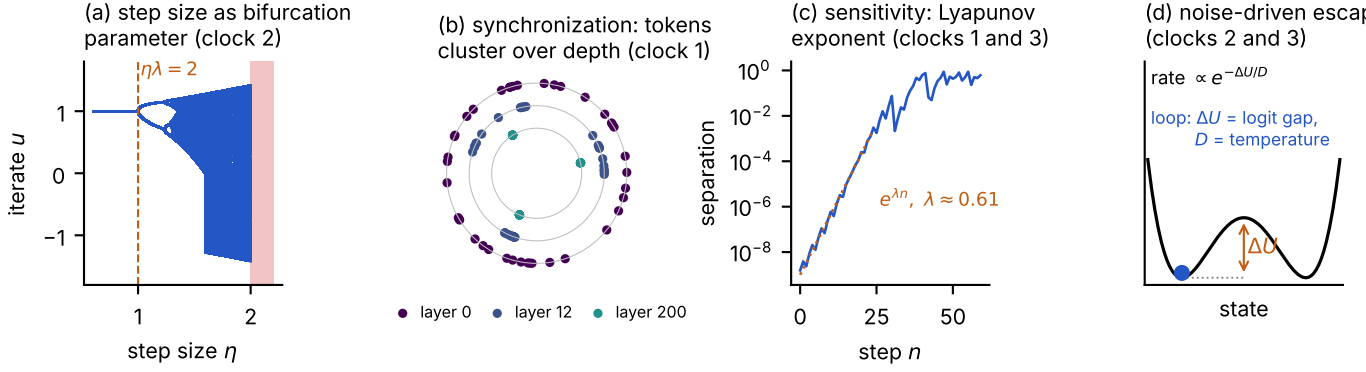

(a) step size as bifurcation parameter (clock 2)
$\eta\lambda = 2$
iterate u
1
0
−1
1
2
step size η
(b) synchronization: tokens cluster over depth (clock 1)
layer 0
layer 12
layer 200
(c) sensitivity: Lyapunov exponent (clocks 1 and 3)
separation
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
$e^{\lambda n},\ \lambda \approx 0.61$
0
25
50
step n
(d) noise-driven escape (clocks 2 and 3)
rate $\propto e^{-\Delta U/D}$
loop: ΔU = logit gap,
D = temperature
ΔU
state